JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CLASS		EXAM INDEX NUMBER	

CHEMISTRY **9647/02**

Paper 2 Structured Questions **14 September 2011**

2 hours

Candidates answer on Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
Total	

1. Planning (P)

You are to plan an experiment to determine the *solubility* of calcium hydroxide, $Ca(OH)_2$, in 0.100 mol dm^{-3} $CaCl_2$ and hence the *solubility product* of calcium hydroxide at the laboratory temperature.

In addition to the standard apparatus present in the laboratory, you are provided with the following materials:

Solid calcium hydroxide, $Ca(OH)_2$
150 cm^3 of 0.100 mol dm^{-3} aqueous calcium chloride, $CaCl_2$
150 cm^3 of 0.200 mol dm^{-3} aqueous hydrochloric acid, HCl
Acid-base indicators
De-ionised water

(a) The pH of a saturated solution of $Ca(OH)_2$ in 0.100 mol dm^{-3} $CaCl_2$ measured using a pH meter was found to be 12.4.

Calculate the concentration of the OH^- ions, in mol dm^{-3}, in the saturated solution, assuming the laboratory temperature is 25°C.

[1]

(b) The concentration of OH^- ions in the saturated solution can also be determined from an acid-base titration.

(i) Using the hydrochloric acid provided and your answer in **(a),** calculate the volume of HCl needed to prepare 250 cm^3 of a diluted HCl solution such that 24.50 cm^3 of the diluted HCl reacts with 25.0 cm^3 of the saturated solution in the titration.

(ii) Briefly describe how you would carry out the dilution to prepare the diluted HCl solution using your answer in **b(i)**.

[2]

1. **(c)** Using volumetric analysis, write a plan to determine the *solubility* of calcium hydroxide, $Ca(OH)_2$ in 0.100 mol dm^{-3} $CaCl_2$ and hence the *solubility product* of $Ca(OH)_2$ at the laboratory temperature.

Your plan should include

- preparation of the saturated solution;
- detailed procedure (including apparatus, appropriate concentration of chemical(s), quantity of chemical, etc);
- measures taken to ensure reliability of result.

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1. **(d)** Assuming that the concentration of the diluted HC*l* used is 0.0245 mol dm^{-3} and 22.75 cm^3 of HC*l* reacted with 25.0 cm^3 of saturated $Ca(OH)_2$ in the above titration, calculate:

(i) the *solubility* of $Ca(OH)_2$, in mol dm^{-3}, in 0.100 mol dm^{-3} $CaCl_2$ at the laboratory temperature.

(ii) the *solubility product* of $Ca(OH)_2$ at the laboratory temperature, stating the units clearly.

[3]

(e) Identify one potential safety hazard in this experiment and suggest how you would minimise this risk.

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.... [1]

[Total: 12]

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2. Group IV elements can form monoxides and dioxides. The stabilities of these monoxides, with respect to disproportionation into the respective elements and the dioxides, vary.

The equations for the disproportionation reactions of some Group IV monoxides are given in the table below, together with some thermodynamic data for the reactions at 298K.

Use of information in the table below is relevant to this question.

Equation	ΔS_r° / J K^{-1} mol^{-1}	ΔH_r° / kJ mol^{-1}	ΔG_r° / kJ mol^{-1}
$2CO(g) \rightarrow C(s) + CO_2(g)$	–178	–173	–120
$2SiO(g) \rightarrow Si(s) + SiO_2(s)$	*to be predicted*	-	–603
$2PbO(s) \rightarrow Pb(s) + PbO_2(s)$	*to be calculated*	+157	+158

(a) Define what is meant by *entropy* of the system.

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…………………………………………………………………………………………….. [1]

(b) Explain the significance of the sign of the standard entropy change, ΔS_r°, for the disproportionation of CO.

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(c) **(i)** Calculate the standard entropy change, ΔS_r°, in J K^{-1} mol^{-1}, for the disproportionation of PbO.

(ii) Comment on the value that you obtained in **(c)(i)**.

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………………………………………………………………………………………. [3]

(d) Hence, predict a value (including sign) for the standard entropy change of the disproportionation of SiO.

Predicted value: ……………………........... J K^{-1} mol^{-1} [1]

2. **(e)** When the disproportionation reaction takes place, $\Delta G_r^{\ominus}$, in J mol^{-1}, is given by the following expression.

$$\Delta G_r^{\ominus} = - RT \ln K_p$$

Calculate the value of the equilibrium constant, K_p, for the disproportionation of CO.

[2]

(f) By using the data provided and data from the *Data Booklet*, construct an energy cycle to calculate the bond energy of C≡O bond in carbon monoxide.

Standard enthalpy change of atomisation of C(s): +715 kJ mol^{-1}

[2]

(g) Suggest why carbon monoxide does not spontaneously disproportionate at room temperature.

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(h) Explain why carbon monoxide is toxic, with reference to its interactions with haemoglobin, an important iron–containing complex in red-blood cells that transports oxygen around the body.

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[Total: 12]

3. Phosphoric acid, H_3PO_4, is often used in teeth whiteners to remove plaque in dentistry.

(a) State the oxidation state of phosphorus in H_3PO_4. …………. [1]

Phosphoric acid is a tribasic acid. The values of the three acid dissociation constants at 25°C of phosphoric acid are given below.

$$H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+ \qquad K_{a1} = 7.50 \times 10^{-3}\ \text{mol dm}^{-3}$$
$$H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+ \qquad K_{a2} = 6.20 \times 10^{-8}\ \text{mol dm}^{-3}$$
$$HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+ \qquad K_{a3} = 4.80 \times 10^{-13}\ \text{mol dm}^{-3}$$

(b) Suggest why there is a decreasing trend in the K_a values.

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…………………………………………………………………………………………... [1]

(c) Identify the conjugate acid of $\mathbf{HPO_4^{2-}}$ and the conjugate base of $\mathbf{HPO_4^{2-}}$.

conjugate acid ……………… conjugate base ……………… [1]

(d) Calculate the K_b of $\mathbf{HPO_4^{2-}}$.

[1]

A phosphate buffer system (made of $\mathbf{H_2PO_4^-}$ and $\mathbf{HPO_4^{2-}}$) serves to buffer the intracellular fluid of cells.

(e) **(i)** Explain what is meant by a *buffer solution*.

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(ii) Write an equation to show how the phosphate buffer system acts as a buffer when a small amount of base is added.

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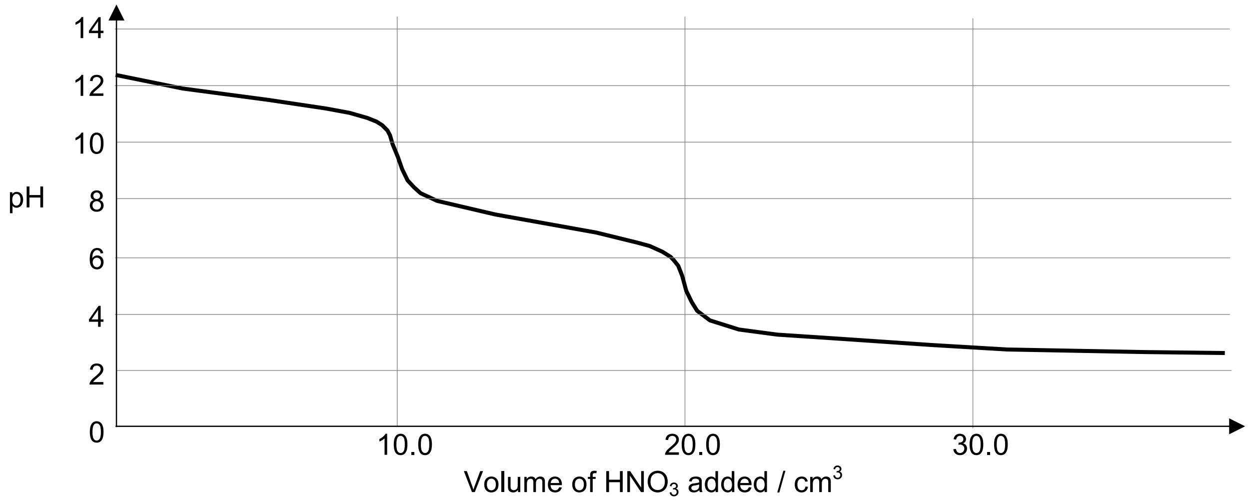
(iii) Using the data provided, calculate the pH of the phosphate buffer when the ratio of [$\mathbf{H_2PO_4^-}$] : [$\mathbf{HPO_4^{2-}}$] is 2 : 1.

[4]

3. The diagram below shows the pH changes when 0.0305 mol dm^{-3} HNO_3 is added to a 25.0 cm^3 solution of sodium phosphate, Na_3PO_4.



(f) Calculate the initial concentration of the sodium phosphate solution.

[1]

(g) State the volume of HNO_3 required to reach the point where the buffer (made of $\mathbf{H_2PO_4^-}$ and $\mathbf{HPO_4^{2-}}$) is operating at its maximum buffering capacity.

…………………… cm^3 [1]

(h) **(i)** From the table below, select an indicator that is most suitable for determining the **second** equivalence point in the titration shown on the graph.

Indicator	**p*K*a**	**acid form**	**base form**
methyl yellow	3.3	red	yellow
bromocresol green	4.9	yellow	blue
chlorophenol red	6.0	yellow	red
bromothymol blue	7.1	yellow	blue
phenolphthalein	9.3	colourless	purple

Indicator: ……………………………………..

(ii) Suggest the colour change observed with this indicator at the end point.

………………………………………………………………………………… [2]

[Total: 12]

4. Like many other transition elements, iron has more than one oxidation state, with +2 and +3 being the most common oxidation states.

(a) Complete the electronic configuration of iron atom and iron(III) ion.

Fe : $1s^2 2s^2 2p^6$…………………………

Fe^{3+} : $1s^2 2s^2 2p^6$………………………… [1]

(b) Explain why iron(III) complexes are often coloured.

[3]

The following reactions involve a sample of aqueous iron(III) nitrate solution.

$$Fe(NO_3)_3(aq) \xrightarrow{NaOH(aq)} Fe(OH)_3(s)$$

$$Fe(NO_3)_3(aq) \xrightarrow{KSCN(aq)} \text{Compound } \mathbf{E} \xrightarrow{KCN(aq)} K_3Fe(CN)_6(aq)$$

(c) **(i)** What is the formula of the cation present in $Fe(NO_3)_3$ (aq) and of the iron-containing cation present in compound **E**?

$Fe(NO_3)_3(aq)$ …………………………….. **E** ……………………………..

(ii) State the colours of $Fe(OH)_3$ and Compound **E**.

$Fe(OH)_3$: ……………………………………………..

Compound **E** : …………………………………………….. [4]

4. **(d)** The presence of CN^- or OH^- significantly affects the oxidising ability of iron(III) ions. Illustrate this effect by quoting appropriate data from the *Data Booklet* and arrange the following three iron(III) compounds in increasing order of oxidising ability.

$Fe(NO_3)_3$, $Fe(OH)_3$, $K_3Fe(CN)_6$

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Oxidising ability: ..……………………………………………………………………….. [2]

(e) Using standard laboratory reagents, describe a simple chemical test to confirm the presence of NO_3^- (aq) ions in a 1 cm^3 sample of solution.

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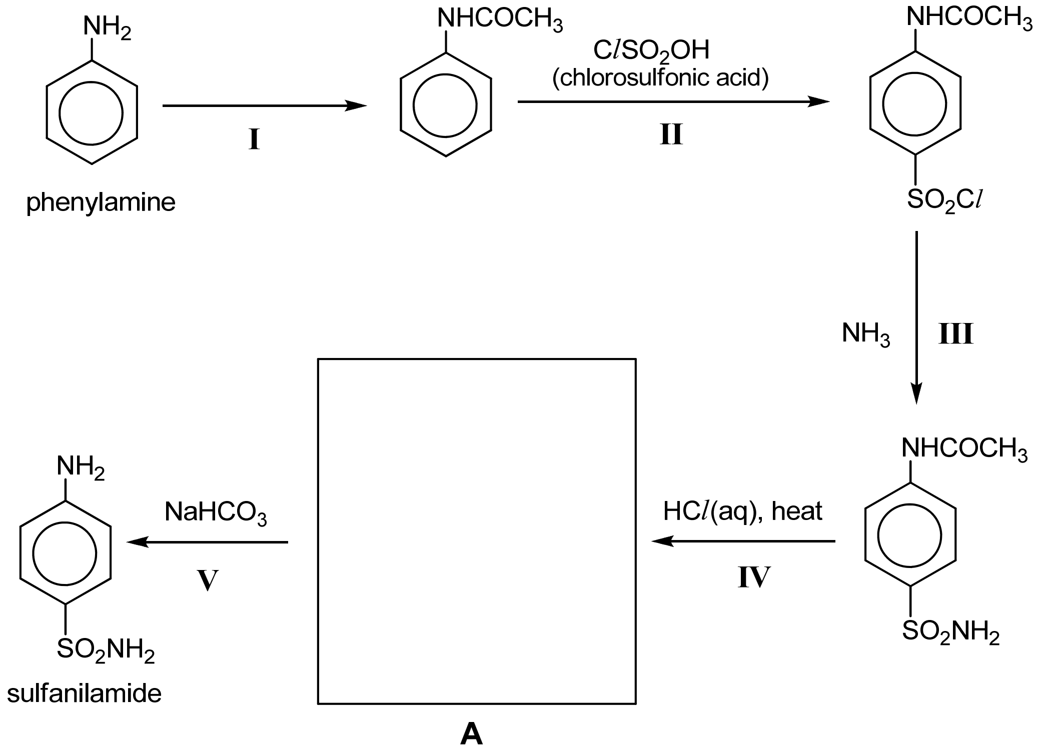
[Total: 11]

5. Sulfanilamide belongs to a class of drugs known as sulfonamides, which are widely used as antibiotics and in the treatment of hair-loss. It can be synthesised from phenylamine via the following synthesis route.



(a) **(i)** State the reagents and conditions required to convert nitrobenzene to phenylamine.

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(ii) State the reagents and conditions for reaction **I**.

…………………………………………………………………………………………

(iii) Name the *type of reaction* occurring in reaction **II**.

………………………………………………………………………………………… [3]

(b) Draw the structural formula of compound **A** in the box provided above. [1]

(c) Suggest why the acetyl group ($-COCH_3$) was introduced in reaction **I** and then removed to regenerate the amine group in sulfanilamide.

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(d) Explain why sulfanilamide is insoluble in water.

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5. **(e)** The basicities of phenylamine, methylamine and trimethylamine are different.

Compound	$C_6H_5NH_2$	CH_3NH_2	$(CH_3)_3N$
pK_b	9.37	3.43	4.20

(i) Arrange the three amines in increasing order of basic strength.

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(ii) Explain the relative basic strength of the three amines in terms of their molecular structures.

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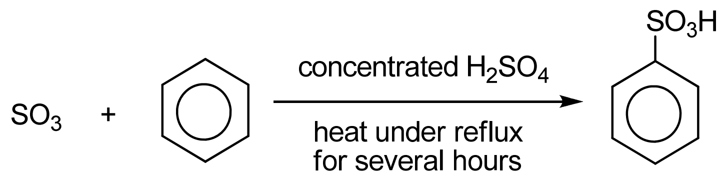
5. (f) The reaction below is an example of an aromatic sulfonation organic reaction in which a hydrogen atom on benzene is replaced by a sulfonic functional group.



In one proposed mechanism for the above reaction, concentrated H_2SO_4 acts as a catalyst. In the first step of the reaction, concentrated H_2SO_4 reacts with SO_3 to generate a cation SO_3H^+.

(i) Write a balanced equation for the first step of the reaction.

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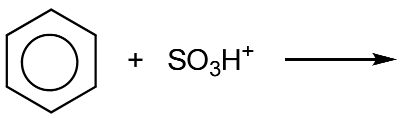
(ii) Complete the mechanism below to describe the subsequent step(s) in the above reaction. Include curly arrows to show the movement of electrons and all charges.



[3]

[Total: 12]

6. Acquired Immuno-Deficiency Syndrome (AIDS) is a disease of the human immune system caused by the Human Immunodeficiency Virus (HIV), which currently infects over 30 million people worldwide.

HIV-1 Protease is an enzyme which speeds up the reproduction of HIV. In the development of anti-HIV drugs, scientists often study the amino acid sequence of the HIV-1 Protease to understand how this enzyme displays its biological properties in the human body.

(a) A segment of the polypeptide structure of HIV-1 Protease, containing 10 amino acids, was digested using an enzyme and the following fragments were obtained.

Leu – Asn – Phe

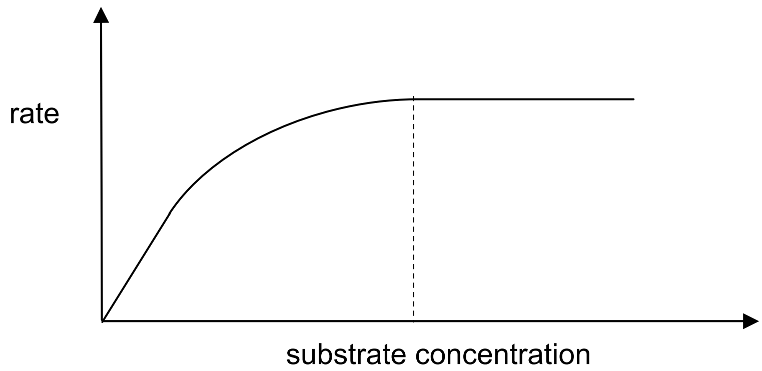
Ile – Gly – Cys – Thr – Leu

Leu – Thr – Glu – Ile – Gly

Deduce the primary structure of this segment of HIV-1 Protease.

………………………………………………………………………………………… [1]

(b) A segment of the HIV strand acts as a substrate which binds to the active site of the HIV-1 Protease which speeds up the reproduction of the HIV virus. The rate of this enzyme-catalysed reaction is investigated and can be represented by the following graph.



(i) How does the graph show that the reaction is zero order with respect to the substrate at high substrate concentration.

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(ii) Explain why it is zero order with respect to the substrate at high substrate concentration.

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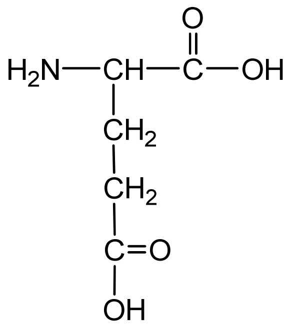
(iii) The catalytic activity of the enzyme can be inhibited by the presence of heavy metal ions, for example, Cu^{2+}.

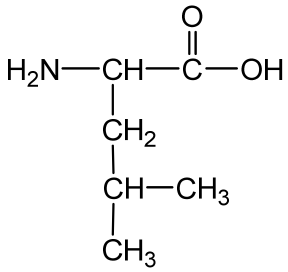
On the same axes above, sketch another graph to show how the rate of reaction is affected when trace amount of heavy metal ions is introduced.

[4]

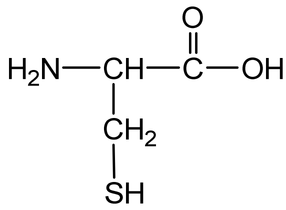
6. **(c)** The structural formulae of the three most common amino acids present in the protein chain in HIV-1 Protease are shown below.



glutamic acid (glu)



leucine (leu)



cysteine (cys)

(d) **(i)** Draw the structural formula of the glu-leu-cys section of the protein chain in HIV-1 Protease at pH 12.

(ii) The side chains of these three amino acids are responsible for maintaining the tertiary structure of HIV-1 Protease.

State the type of side-chain interaction between two cysteine (cys) residues. Illustrate your answer with a suitable diagram.

Side-chain interaction : ……………………………………

6. **(d)** **(iii)** With reference to structures of glutamic acid (glu) and leucine (leu), describe what is meant by the *quaternary structure* of *HIV-1 Protease*. In your answer, you should state the type of bonding or interaction involved.

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(e) Since its discovery in 1981, scientists have studied the protein structure of HIV in order to prevent its transmission., The World Health Organisation recommends heat treatment of breast-milk prior to giving it to the child, as a way to reduce the risk of HIV transmission to the baby in sub-saharan Africa

Explain how this treatment of breast-milk reduces the risk of HIV infection in infants.

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[Total: 13]

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